

have a higher binding energy per nucleon.

- 6 (a) (i) Gravitational potential at a point in the gravitational field is the work done per unit mass in bringing the mass from infinity to that point.
- (ii) Gravitational force is an attractive force and the force acting on a test mass is pointing in a direction towards the point mass m . The external force, which is opposite to the gravitational force, does negative work in bringing the test mass from infinity to a point r from the point mass. This is because the external force and the displacement are pointing in opposite directions. Hence, gravitational potential at that point is negative.
- (b) (i) For the satellite to be in orbit with speed v , gravitational force must provide the centripetal force. By Newton's second law of motion,

$$F_g = ma_c$$

$$\frac{GMm}{R^2} = \frac{mv^2}{R}$$

$$\frac{1}{2} \frac{GMm}{R} = \frac{1}{2} mv^2$$

Since kinetic energy of the satellite is $\frac{1}{2}mv^2$, $E_k = \frac{GMm}{2R}$.

Note: Derivation should be accompanied by short statements illustrating the physics of the situation. A statement where gravitational force provides the centripetal force for the satellite to move in circle is expected from candidates.

(ii) The potential energy of the satellite is $U = -\frac{GMm}{R}$.

Hence, the total energy of the satellite is

$$E_T = E_k + U$$

$$E_T = \frac{GMm}{2R} + \left(-\frac{GMm}{R}\right)$$

$$E_T = -\frac{GMm}{2R} \text{ (shown)}$$

Note: There should be clear explanations of the physics concepts involved, rather than just showing the mathematical manipulation.

(c) (i) From (b)(i), we have $\frac{1}{2}mv^2 = \frac{1}{2}\frac{GMm}{R}$. Therefore,

$$v = \sqrt{\frac{GM}{R}}$$

$$v = \sqrt{\frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{7.2 \times 10^6}}$$

$$v \approx 7.46 \times 10^3 \text{ ms}^{-1}$$

(ii) By Newton's second law of motion, we have

$$F_g = ma_c$$

$$\frac{GMm}{R^2} = mR\omega^2$$

$$T^2 = \frac{4\pi^2}{GM}R^3$$

$$T^2 = \frac{4\pi^2}{6.67 \times 10^{-11} \times 6.0 \times 10^{24}} \times (7.2 \times 10^6)^3$$

$$T \approx 1.69 \text{ hours}$$

Since the period of revolution is not equal to 24 hours, this means that the satellite is not in a geostationary orbit.

(d) As satellite loses total energy, its orbital radius decreases and it falls back closer to the Earth. At the same time, its orbital speed increases since its orbital radius decreases. Since the satellite is travelling at high speed and continuously loses total energy due to increasing resistive forces in the Earth's atmosphere, there will be an increasing rate of conversion of energy to heat, which is why satellite 'burn up'.

7 (a) Simple harmonic motion is an oscillatory motion whereby the acceleration of a body